MATH441/MATH541
Applied Mathematics II: Complex Variables
Lecture slides 17: Isolated Singularities, Poles, Residue Theorem

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Lecture outline

1. Isolated Singularities
2. Classification of Singularities
3. Pole of order m
4. The Residue Theorem

Laurent Series Theorem - Review

Theorem

Laurent Expansion *Let $0 \leq r < R \leq \infty$ and suppose f is differentiable the annulus $r < |z - z_0| < R$. Then for each z in this annulus,*

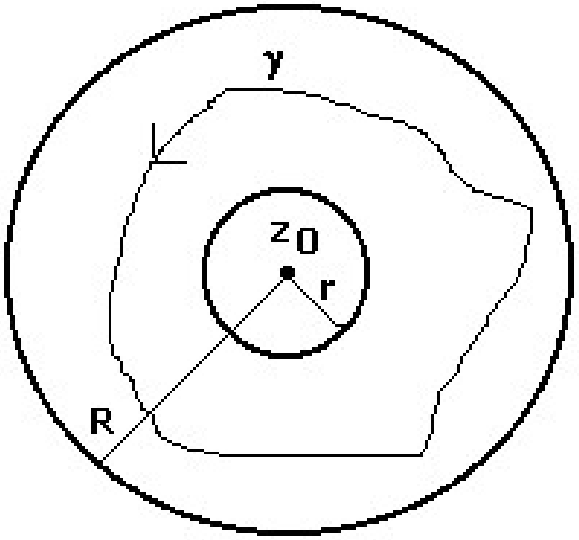
$$f(z) = \sum_{n=-\infty}^{\infty} c_n (z - z_0)^n \qquad (1)$$

where for each integer n

$$c_n = \frac{1}{2\pi i} \oint_\gamma \frac{f(z)}{(z - z_0)^{n+1}} dz \qquad (2)$$

where γ is any closed path about z_0 lying entirely in the annulus.

Laurent Series Theorem - Review. Part 2



Remark: Just as in the case of Taylor series, one rarely finds the Laurent expansion of a function using the expression given by (2) . In fact, we shall actually see later that one can use one of these coefficients to evaluate integrals. Usually, one uses a known series to generate other series. This can be done since the next theorem assures us that the Laurent expansion of a function about a point is unique.

Isolated Singularities

Suppose $f(z)$ is regular and single-valued for $0 < |z - z_0| < r$. Then we have the expansion

$$f(z) = \cdots + \frac{a_{-2}}{(z - z_0)^2} + \frac{a_{-1}}{(z - z_0)} + a_0 + a_1 (z - z_0) + a_2 (z - z_0)^2 + \cdots \qquad (1)$$

which is convergent for $0 < |z - z_0| < r$. If all the a_{-n} $(n = 1, 2, \ldots)$ are zero, then (1) reduces to a Taylor series convergent for $0 \leq |z - z_0| < r$, i.e., $f(z)$ is regular at $z = z_0$. If not all of the a_{-n} are zero, then $z = z_0$ is a singular point for $f(z)$. We say in this case that $z = z_0$ is an *isolated singularity* of $f(z)$. Thus

Definition

A complex function f has an *isolated singularity* at z_0 if f is differentiable in an annulus $0 < |z - z_0| < r$ but not at z_0 itself.

Isolated Singularities: examples

Example

$f(z) = \dfrac{1}{\sin\left(\frac{1}{z}\right)}$ has a singularity at $z = 0$ which is not isolated, since $\sin\left(\frac{1}{z}\right) = 0$ at $\frac{1}{z} = n\pi \quad (n = 0, \pm 1, \pm 2, \ldots)$ Thus $\sin\left(\frac{1}{z}\right) = 0$ for $z = \frac{1}{n\pi}$.

Example

$f(z) = \frac{1}{\sin z}$ has an isolated singularity at $z = 0$.

Classification of Singularities

Definition

Classification of Singularities Let f have a singularity at z_0. Then in some annulus $0 < |z - z_0| < r$, $f(z)$ has a Laurent expansion of the form

$$f(z) = \sum_{n=-\infty}^{\infty} c_n (z - z_0)^n .$$

We classify the singularity z_0 as follows:

1. z_0 is a *removable singularity* if $c_n = 0$ for $n = -1, -2, -3, \ldots$;
2. z_0 is a *pole of order* m (m a positive integer), if $c_{-m} \neq 0$, but $c_{-m-1} = c_{-m-2} = \cdots = 0$;
3. z_0 is an *essential singularity* if $c_{-n} \neq 0$ for infinitely many positive integers n.

Classification of Singularities. Part 2

Remark: In the case of a removable singularity at z_0 the Laurent expansion about z_0 does not have any negative powers of $z - z_0$. Thus the expansion is of the form $f(z) = \sum_{n=0}^{\infty} c_n (z - z_0)^n$ in some annulus $0 < |z - z_0| < r$. But this is actually a Taylor series having the value c_0 at z_0. This where the term removable comes from. By defining $f(z) = c_0$ we get a function that is differentiable at z_0, "removing" the singularity.

Example

Let $f(z) = \dfrac{\sin z}{z}$ for $z \neq 0$. Then $z = 0$ is a removable singularity since

$$\frac{\sin z}{z} = \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n+1)!} z^{2n} = 1 - \frac{1}{6} z^2 + \frac{1}{120} z^4 + \cdots$$

for $z \neq 0$, and this series has no terms involving $\frac{1}{z}$. If we define $f(0) = 1$, then we remove the singularity and extend f to a function that is differentiable at 0.

Classification of Singularities. Part 3

Remark: If $f(z)$ has a pole of order m at z_0, then

$$f(z) = \frac{c_{-m}}{(z-z_0)^m} + \frac{c_{-m+1}}{(z-z_0)^{m-1}} + \cdots + \frac{c_{-1}}{z-z_0} + c_0 + c_1(z-z_0) + \cdots$$

where $c_{-m} \neq 0$. The rational function

$$\frac{c_{-m}}{(z-z_0)^m} + \frac{c_{-m+1}}{(z-z_0)^{m-1}} + \cdots + \frac{c_{-1}}{z-z_0}$$

is called the *principal part* or *meromorphic part* of $f(z)$ at $z = z_0$.

Definition

A pole of order 1 is called a *simple pole*. A pole of order 2 is called a *double pole*.

Example

$\dfrac{z^2 - z}{(z+1)^2(z^2+4)}$ has a double pole at $z = -1$ and simple poles at $z = \pm 2i$.

Condition for a pole of order m. Part 1.

Theorem

Condition for a pole of order m Let f be differentiable in the annulus $0 < |z - z_0| < r$. Then f has a pole of order m at z_0 if and only if

$$\lim_{z \to z_0} (z - z_0)^m f(z)$$

is a nonzero constant.

Condition for a pole of order m. Part 2

Proof: Expanding f in a Laurent series in the annulus we have

$$f(z) = \sum_{-\infty}^{\infty} c_n (z - z_0)^n .$$

Suppose that f has a pole of order m at z_0. Then $c_{-n} = 0$ for $n = m+1, m+2, \ldots$ Thus

$$(z - z_0)^m f(z) = c_{-m} + c_{-m+1}(z - z_0) + c_{-m+2}(z - z_0)^2 + \cdots$$

so that

$$\lim_{z \to z_0} (z - z_0)^m f(z) = c_{-m} \neq 0.$$

Condition for a pole of order m. Part 3.

Conversely, suppose $\lim_{z\to z_0}(z-z_0)^m f(z) = L \neq 0$. We must show that f has a pole of order m at z_0.
Now

$$f(z) = \sum_{-\infty}^{\infty} c_n (z-z_0)^n$$

in some annulus about z_0,where

$$c_n = \frac{1}{2\pi i}\oint_\gamma \frac{f(z)}{(z-z_0)^{n+1}}dz$$

and γ is a closed curve enclosing z_0 in this annulus. Let $\varepsilon > 0$ be given. Then $\exists$ $\delta > 0$ such that

$$\left|(z-z_0)^m f(z) - L\right| < \varepsilon \qquad \text{if } 0 < |z-z_0| < \delta$$

But $\left|\left|(z-z_0)^m f(z)\right| - |-L|\right| \leq \left|(z-z_0)^m f(z) - L\right| < \varepsilon$ since $||z|-|w|| \leq |z+w|$ and this implies that $||z|-|w|| = ||z|-|-w|| \leq |z-w|$.

Condition for a pole of order m. Part 4.

Thus

$$\left|(z-z_0)^m f(z)\right| < |L| + \varepsilon \qquad \text{if } 0 < |z-z_0| < \delta$$

Taking $|z-z_0| = \delta$ gives

$$\left|(z-z_0)^{-n-1} f(z)\right| < (|L|+\varepsilon)\left|(z-z_0)^{-n-m-1}\right| = (|L|+\varepsilon)\,\delta^{-n-m-1}$$

Hence

$$|c_n| \leq \frac{1}{2\pi}\oint_\gamma \left|\frac{f(z)}{(z-z_0)^{n+1}}\right| dz \leq \frac{1}{2\pi}(|L|+\varepsilon)\,\delta^{-n-m-1}(2\pi\delta) = (|L|+\varepsilon)\,\delta^{-n-m}$$

where we have used the deformation theorem to replace γ by the circle $|z-z_0| = \delta$. If $n < -m$, then $0 < -n-m$ and we can make δ^{-n-m} as small as we like by choosing δ sufficiently small. Thus $c_n = 0$ if $n < -m$. Also,

$$\lim_{z\to z_0}(z-z_0)^m f(z) = c_{-m} = L \neq 0.$$

Thus f has a pole of order m at z_0.

Condition for a pole of order m. Part 5.

Example

Let $f(z) = \dfrac{1}{z+i}$. Then $\lim_{z \to -i} (z+i) f(z) = 1 \neq 0$ so this function has a simple pole at $-i$.

Remark: Poles are intimately related to zeroes of a function, because often a quotient has a pole where the denominator vanishes. Recall that z_0 is called a *zero* of f if $f(z_0) = 0$. If f is differentiable in an open disk about z_0, then the *order* of the zero in the index of the first nonzero coefficient in the Taylor series expansion of f about z_0.

Condition for a pole of order m. Part 6.

Thus f has a zero of order m when it is of the form

$$f(z) = \sum_{n=m}^{\infty} c_n (z - z_0)^n = c_m (z - z_0)^m + c_{m+1} (z - z_0)^{m+1} + \cdots$$

$c_m \neq 0$. From the Taylor formula for these coefficients it follows that this is equivalent to the condition

$$f(z_0) = f'(z_0) = \cdots = f^{(m-1)}(z_0) = 0, \qquad \text{but } f^{(m)}(z_0) \neq 0$$

Example

Thus $f(z) = (z + 3i)^3$ has a third order zero at $-2i$ since $f(-2i) = f'(-2i) = f''(2i) = 0$ but $f^{(3)}(3i) \neq 0$.

Condition for a pole of order m. Part 7.

Theorem

Let

$$f(z) = \frac{h(z)}{g(z)}$$

where h and g are differentiable in an open disk about z_0. Suppose $h(z_0) \neq 0$ and g has a zero or order m at z_0. Then f has a pole of order m at z_0.

Theorem

Poles of Products Let f have a pole of order m at z_0 and let g have a pole of order n at z_0. Then fg has a pole of order $m+n$ at z_0.

Theorem

Let g have a pole of order m at z_0 and f have a zero or order $n < m$ at z_0. Then fg has a pole of order $m-n$ at z_0.

The Residue Theorem. Part 1.

Definition

Let f have an isolated singularity at z_0. In some annulus $0 < |z - z_0| < R$, f has a Laurent expansion

$$f(z) = \sum_{n=-\infty}^{\infty} c_n (z - z_0)^n .$$

The number c_{-1} is called the *residue* of f at z_0, and is denoted by $\mathrm{Res}(f, z_0)$.

Thus the residue of f at z_0 is the coefficient of $1/(z - z_0)$ in the Laurent expansion of f. From Laurent's theorem

$$c_n = \frac{1}{2\pi i} \oint_\gamma \frac{f(z)}{(z - z_0)^{n+1}} dz$$

where γ is a positively oriented closed path about z_0 in the annulus $0 < |z - z_0| < R$.

The Residue Theorem. Part 2.

Hence

$$c_{-1} = Res\left(f, z_0\right) = \frac{1}{2\pi i} \oint_{\gamma} f\left(z\right) dz.$$

Therefore,

$$\oint_{\gamma} f\left(z\right) dz. = 2\pi i \, Res\left(f, z_0\right).$$

Hence, knowing the residue of f at z_0 is equivalent to evaluating the integral above.

The conclusion just stated is limited since we have required that γ enclose only one singularity of f. However, the residue theorem extends this result to the case where γ encloses any finite number of singularities of f.

Thank you